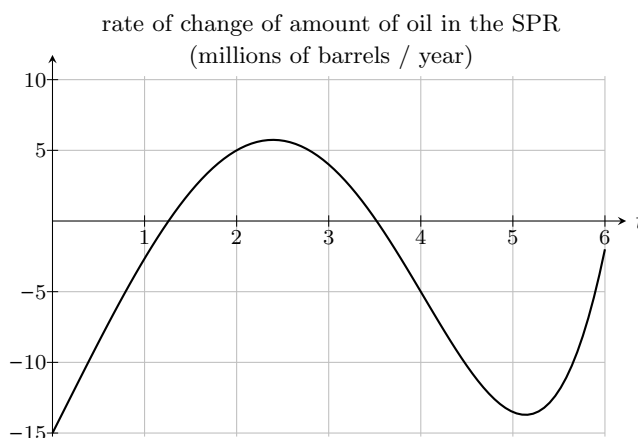


Net Change/ The Definite Integral

1

The Strategic Petroleum Reserve (SPR) is an emergency supply of oil maintained by the United States Department of Energy. The graph below is a model of $f(t)$, the *rate* at which oil enters or leaves the reserve in millions of barrels per year (a positive rate means oil is entering; a negative rate means oil is leaving). Time t is measured in years.



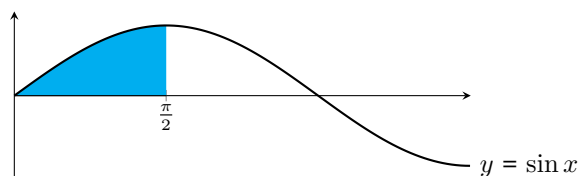
We'd like to approximate the net change in the amount of oil in the SPR between $t = 0$ and $t = 6$. As you learned in class, the idea is to pretend that the rate of change is constant for short time intervals.

- (a) At what rate was oil entering/leaving the SPR at $t = 0$? (For this part and all following, give your answer *with units*, and be sure to say whether the oil was entering or leaving.)
- (b) If we *pretended*, for the purposes of simplicity, that oil entered/left the reserve at a *constant* rate (namely, the rate you found in (a)) between $t = 0$ and $t = 2$, how much oil entered/left in that time period?
- (c) At what rate was oil entering/leaving the reserve at $t = 2$?
- (d) If we pretended that oil entered/left the reserve at this same rate between $t = 2$ and $t = 4$, how much oil entered/left in that time period?
- (e) At what rate is oil entering/leaving the reserve at $t = 4$?

- (f) If we pretended that oil entered/left the reserve at this same rate between $t = 4$ and $t = 6$, how much oil entered/left in that time period?
- (g) Use your answers to (b), (d), and (f) to estimate the net change in the amount of oil from $t = 0$ to $t = 6$. Was this a net gain or a net loss?
- (h) Illustrate your answer to (g) on the graph of f . Did you do a right-hand or left-hand approximation?

2

In this problem, we are interested in the area between the graph of $\sin x$ and the x -axis for x in the interval $[0, \frac{\pi}{2}]$. (This is the shaded area in the graph below.)



- (a) Recall that we use the notation L_3 to mean the left-hand sum with three rectangles (of equal width) that estimates this area. Find L_3 ; give an exact answer. Draw a picture illustrating L_3 . Is L_3 an overestimate or underestimate for the area we're interested in?

- (b) Repeat (a) using a right-hand sum rather than a left-hand sum.

- (c) Use the Riemann Sums¹ applet at <https://www.geogebra.org/m/MYn6MgTQ> to find L_{50} and R_{50} . (You can also use it to check your answers for L_3 and R_3 .) Based on these, what do you think the actual area we're looking for might be?

To use the applet, type `sin(x)` in the box for $f(x)$, 0 in the box for a , and `pi/2` in the box for b ; and $n = 50$. This will give you a left-hand sum, a right-hand sum, and something called a midpoint sum which is (not surprisingly) somewhere in between.

- (d) When you estimate the area under the curve $y = \sin(x)$ between 0 and $\frac{\pi}{2}$ using 50 rectangles, what is the width of each rectangle in your figure?

¹“Riemann sum” is just another way of saying “left-hand sum or right-hand sum”. These sums are named for Bernhard Riemann, a 19th century mathematician.

- (e) When you estimate the area under the curve $y = \sin(x)$ between 0 and $\frac{\pi}{2}$ using n rectangles, what is the width of each rectangle in your figure? Give your answer in terms of n and justify your reasoning.

3

Reflect Back. What characteristic of f will ensure that a right-hand sum gives an underestimate of the signed area between the graph of f and the x -axis? Does it matter whether f is positive or negative? Increasing or decreasing? Concave up or concave down?

Illustrate your answer with some pictures.

4

Evaluate the following definite integrals by interpreting each as a signed area. It will be very helpful to draw a graph.

(a) $\int_{-1}^3 3 \, dx$

(b) $\int_{-2}^5 |x - 2| \, dx$

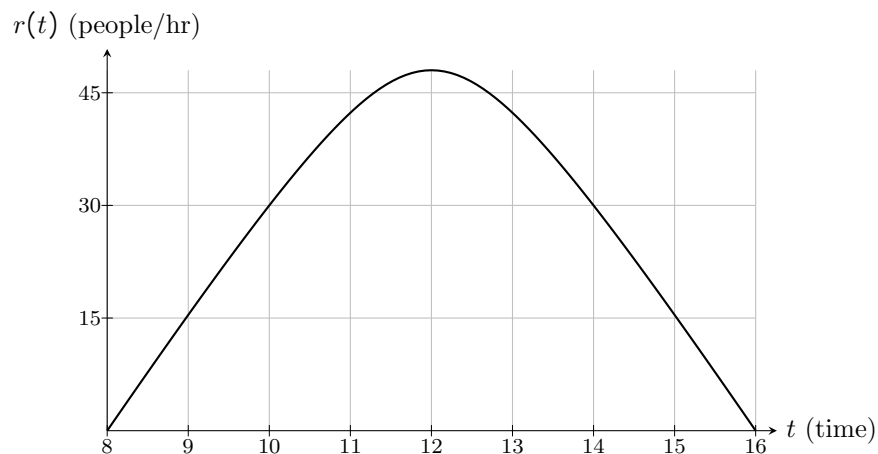
(c) $\int_{-2}^5 (|x| - 2) \, dx$

(d) $\int_{-3}^3 \arctan(x) \, dx$

(e) $\int_{-3}^3 \sqrt{9 - x^2} \, dx$

5

A health clinic has set up a flu shot program for its patients. The clinic is open from 8 am to 4 pm (16:00) and gives flu shots on a first-come, first-serve basis. The clinic has the capacity to serve 30 patients per hour. The function $r(t)$, whose graph is given below, gives the rate at which people are arriving at the clinic for shots one day.



Explain your answers to the questions below by relating them to points, lengths, or areas on the graph.

- (a) If someone arrives at 9 AM, how long do they have to wait for a shot?
- (b) At what time does a line start forming?
- (c) At approximately what time is the length of the line increasing most rapidly?
- (d) At approximately what time is the line the longest? (The answers to this is different from the answer to (c); explain why.)
- (e) When the line is longest, approximately how many people are in line?
- (f) Approximate the longest amount of time a person could wait for a shot. (Assume that doors close to new arrivals at 4 pm but everyone who has arrived by 4 pm is served.)
- (g) Approximately how long is the line at 3 pm?
- (h) Approximate the number of people who came to the clinic for a flu shot this day.

6

For the scenario described in #5, express the following quantities exactly; your answers may be in terms of definite integrals. (For instance, your answer might look like $50 + \int_8^{10} r(t) dt$ or $\int_8^9 [r(t) - 2] dt$.)

- (a) The length of the line at 11 am
- (b) The length of the line at 3 pm
- (c) The amount of time a person arriving at noon has to wait in line
- (d) The amount of time by which the clinic must extend its hours in order to serve everyone who arrives by 4pm